

Fig. 4: Sort 1D Array function



Fig. 5: Array-to-Cluster function



Fig. 6: Unbundle-by-Name function

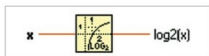


Fig. 7: Logarithm Base 2 function

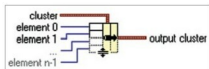


Fig. 8: Bundle function



Fig. 9: Quotient and Remainder function



Fig. 10: Clusters

comparing the first elements. If the first elements match, the function compares the second and subsequent elements.

The connector pane displays default data types for this polymorphic function.

Array-to-Cluster function. It converts a 1D array to a cluster of elements of the same type as the array elements as shown in Fig. 5. The array is a 1D array of any type and each cluster element is the same as the corresponding element in the array. Cluster order matches the order of elements in the array.

The default number of elements in the cluster is nine and the maximum cluster size for this function is 256. Use this function when you want to display elements of the same type in a front-panel cluster indicator and also want to manipulate the elements on the block diagram by their index values.

Unbundle-by-Name function. It returns the cluster elements whose names you specify as shown in Fig. 6. Input cluster is the cluster whose elements you want to access and element 0..m-1 is the element of input cluster called 0..m-1.

We do not have to keep track of the order of the elements within the cluster. This function does not require the number of elements to match the number in the cluster. After you wire a cluster to this function, you can select individual elements from the function.

Logarithm Base 2 function. It computes base 2 logarithm of x . If x is 0, $\log_2(x)$ is negative infinity as shown in Fig. 7. Here, x can be a scalar number, array or cluster of numbers, array of clusters of numbers and so on. $\log_2(x)$ is of the same numeric representation as x .

Bundle function. If x is not complex and is less than 0, $\log_2(x)$ is NaN (not a number).

It assembles a cluster from individual elements as shown in Fig. 8. The cluster is one whose value you want to change and element 0..n-1 accepts any data type, and output cluster is the resulting cluster.

You can also use this function to change the values of individual elements in an existing cluster without having to specify new values for all elements. To do so, wire the cluster you want to change to the middle cluster terminal of this function. When you wire a cluster to this function, function resizes automatically to display inputs for each element in the cluster.

Quotient and Remainder function. Computes integer quotient and remainder of the two inputs as shown in Fig. 9. x and y can be scalar numbers; $x-y*\text{floor}(x/y)$ is the remainder and $\text{floor}(x/y)$ is the integer quotient. This function rounds $\text{floor}(x/y)$ to the nearest integer towards infinity.

Clusters. Clusters group data elements of mixed types as shown in

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Fig. 10. A cluster is similar to a record or struct in text based programming languages. An example of a cluster is LabVIEW error cluster, which combines a Boolean value, a numeric value and a string.

Testing procedure

1. Install LabVIEW
2. Click LabVIEW on the desktop
3. Open Shannoncode.vi
4. Provide source symbols a0..a3 with probabilities so that total probability is equal to one
5. Click Operate → Run
Generated codes should satisfy the prefix and instantaneous properties

Bundling several data elements into clusters eliminates wire clutter on the block diagram and reduces the number of connector pane terminals that subVIs need. The connector pane has, at most, 28 terminals. If your front panel contains more than 28 controls and indicators that you want to pass to another VI, group some of these into a cluster and assign the same to a terminal on the connector pane.

Most clusters on the block diagram have a pink wire pattern and data-type terminal. Error clusters have a dark-yellow wire pattern and data-type terminal. Cluster elements have a logical order unrelated to their position in the shell. The first object you place in the cluster is element 0, second is element 1 and so on. If you delete an element, the order adjusts automatically. The cluster order determines the order in which the elements appear as terminals on Bundle and Unbundle functions on the block diagram. **EFY**

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